# WS-01: Test Card TC-01 — Ground Bench Test Worksheet

## STATIC THRUST, ELECTRICAL POWER & THERMAL EQUILIBRIUM (RIGID LOAD STAND)

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|---|---|---|---|
| **AIRCRAFT:** MCO-1D Mark 1 | **CARD ID:** TC-01 | **DATE:** ____ / ____ / 2026 | **LOCATION:** _____ |
| **OPERATOR / TECH:** _____ | **AIRFRAME S/N:** MCO-1D-001 | **AMBIENT TEMP:** _____ °C | **BARO / QNH:** _____ hPa |

### 1. Flapping Frequency Sweep & Static Performance Matrix

Mount airframe securely to rigid load stand. Execute frequency steps sequentially. Record peak readings during final 10s of dwell.

| Pt | Cmd Freq | Stroke Amp | Dwell | Target Thrust | Measured Thrust | GS Bus Current | Peak Motor Temp | ODrive FET Temp | Cable Slip OK? |
|---|---|---|---|---|---|---|---|---|---|
| **P1** | 1.0 Hz | ± 60.0° | 30 s | *≥ 8.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |
| **P2** | 1.5 Hz | ± 70.0° | 30 s | *≥ 16.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |
| **P3** | 2.0 Hz | ± 80.0° | 30 s | *≥ 24.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |
| **P4** | 2.5 Hz | ± 80.0° | 45 s | *≥ 31.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |
| **P5** | 3.0 Hz | ± 80.0° | 60 s | *≥ 37.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |
| **P6** | 3.5 Hz | ± 70.0° | 20 s | *≥ 42.0 N* | [ ] N | [ ] A | [ ] °C | [ ] °C | [ ] OK |

### 2. Electrical Subsystem, Servo Rail & Structural Dynamics Checks

| Check / Parameter Description | Sensor / Measuring Point | Target Pass Criteria | Measured Value | Evaluation |
|---|---|---|---|---|
| 7.4V Servo Bus Rail Loaded Voltage | Oscilloscope across 7.4V rail at P5 | *≥ 7.10 V (droop ≤ 0.3 V)* | [ ] V | [ ] PASS [ ] FAIL |
| 16-Servo Simultaneous Reversal Surge | Current clamp on 2S battery lead | *≤ 45.0 A peak transient* | [ ] A | [ ] PASS [ ] FAIL |
| 6S Main Bus Voltage at Full 3.5 Hz Load | ODrive vbus telemetry | *≥ 21.60 V (no severe sag)* | [ ] V | [ ] PASS [ ] FAIL |
| Dyneema Max Angular Slippage | Witness mark alignment on sector rim | *≤ 0.5° (zero visible shift)* | [ ] ° | [ ] PASS [ ] FAIL |
| Airframe RMS Accelerometer Vibration | Pixhawk 4 Mini IMU log (10–200 Hz) | *≤ 2.0 g RMS* | [ ] g | [ ] PASS [ ] FAIL |
| Wing Spar Deflection at Root under P6 | Optical deflection scale at station yoke | *No delamination, crack, or slip* | [ ] | [ ] PASS [ ] FAIL |

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|---|---|
| **GATE CERTIFICATION: TC-01 GROUND BENCH TEST QUALIFICATION** | **STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST** |
| **INSPECTOR / LEAD:** _____ | **SIGNATURE & DATE:** _____ |
| **REMARKS / DISCREPANCIES:** _____ | |